

External Table LAB 5

```
USE DATABASE HRMS;
```

```
USE SCHEMA HR;
```

```
CREATE OR REPLACE FILE FORMAT HRMS.HR.JSON_ET_FILEFORMAT  
TYPE = JSON;
```

```
CREATE OR REPLACE STAGE HRMS.HR.AWS_ET_JSON_STAGE  
STORAGE_INTEGRATION = AWS_S3_INT  
URL = 's3://learn2cloud-snowflake/external_table/JSON/'  
FILE_FORMAT = JSON_ET_FILEFORMAT;
```

```
LIST @AWS_ET_JSON_STAGE;
```

```
SELECT * FROM @AWS_ET_JSON_STAGE;
```

```
SELECT $1 FROM @AWS_ET_JSON_STAGE;
```

```
CREATE OR REPLACE EXTERNAL TABLE EXT_EMP_JSON  
(  
  EMP_JSON_DATA VARIANT AS (VALUE::VARIANT)  
)  
WITH LOCATION = @AWS_ET_JSON_STAGE  
FILE_FORMAT = JSON_ET_FILEFORMAT;
```

```
SELECT * FROM EXT_EMP_JSON;
```

```
CREATE OR REPLACE EXTERNAL TABLE EXT_EMP_JSON_1  
(  
  EMP_JSON_DATA VARIANT AS (VALUE::VARIANT),  
  EMP_ID VARCHAR AS ($1[0].EMPLOYEE_ID::VARCHAR)  
)  
WITH LOCATION = @AWS_ET_JSON_STAGE
```

```
FILE_FORMAT = JSON_ET_FILEFORMAT;
```

```
SELECT * FROM EXT_EMP_JSON_1;
```

```
CREATE OR REPLACE EXTERNAL TABLE EXT_EMP_JSON_2  
(  
  EMP_JSON_DATA VARIANT AS (VALUE::VARIANT),  
  EMP_ID VARCHAR AS ($1[0].EMPLOYEE_ID::VARCHAR),  
  EMP_ID2 VARCHAR AS ($1[1].EMPLOYEE_ID::VARCHAR)  
)  
WITH LOCATION = @AWS_ET_JSON_STAGE  
FILE_FORMAT = JSON_ET_FILEFORMAT;
```

```
SELECT * FROM EXT_EMP_JSON_2;
```

```
SELECT DISTINCT METADATA$FILENAME FROM EXT_EMP_JSON;
```

```
SELECT * FROM EXT_EMP_JSON EDJ,  
lateral flatten( input =>EMP_JSON_DATA) Y;
```

```
SELECT  
  Y.VALUE:EMPLOYEE_ID,  
  Y.VALUE:FIRST_NAME,  
  Y.VALUE:LAST_NAME,  
  Y.VALUE:EMAIL,  
  Y.VALUE:PHONE_NUMBER,  
  Y.VALUE:HIRE_DATE,  
  Y.VALUE:JOB_ID,  
  Y.VALUE:SALARY,  
  Y.VALUE:COMMISSION_PCT,  
  Y.VALUE:MANAGER_ID,  
  Y.VALUE:DEPARTMENT_ID  
FROM EXT_EMP_JSON EDJ,  
lateral flatten( input =>EMP_JSON_DATA) Y;
```

```
SELECT  
  Y.VALUE:EMPLOYEE_ID:: INTEGER,  
  Y.VALUE:FIRST_NAME :: VARCHAR,
```

```

Y.VALUE:LAST_NAME :: VARCHAR,
Y.VALUE:EMAIL:: VARCHAR,
Y.VALUE:PHONE_NUMBER:: VARCHAR ,
Y.VALUE:HIRE_DATE:: VARCHAR,
Y.VALUE:JOB_ID:: VARCHAR,
Y.VALUE:SALARY:: VARCHAR,
Y.VALUE:COMMISSION_PCT:: VARCHAR,
Y.VALUE:MANAGER_ID:: VARCHAR,
Y.VALUE:DEPARTMENT_ID:: VARCHAR
FROM EXT_EMP_JSON EDJ,
lateral flatten( input =>EMP_JSON_DATA) Y;

```

```

CREATE OR REPLACE VIEW VIEW_EXT_EMP_JSON
AS
SELECT
    Y.VALUE:EMPLOYEE_ID:: INTEGER    AS EMPLOYEE_ID,
    Y.VALUE:FIRST_NAME :: VARCHAR    AS FIRST_NAME ,
    Y.VALUE:LAST_NAME :: VARCHAR     AS LAST_NAME,
    Y.VALUE:EMAIL:: VARCHAR          AS EMAIL,
    Y.VALUE:PHONE_NUMBER:: VARCHAR   AS PHONE_NUMBER,
    Y.VALUE:HIRE_DATE:: VARCHAR      AS HIRE_DATE,
    Y.VALUE:JOB_ID:: VARCHAR         AS JOB_ID,
    Y.VALUE:SALARY:: VARCHAR         AS SALARY,
    Y.VALUE:COMMISSION_PCT:: VARCHAR AS COMMISSION_PCT,
    Y.VALUE:MANAGER_ID:: VARCHAR     AS MANAGER_ID,
    Y.VALUE:DEPARTMENT_ID:: VARCHAR  AS DEPARTMENT_ID
FROM EXT_EMP_JSON EDJ,
lateral flatten( input =>EMP_JSON_DATA) Y;

```

```

SELECT * FROM VIEW_EXT_EMP_JSON;

```

```
SELECT * FROM VIEW_EXT_EMP_DEPT_JSON;
```

```
SELECT * FROM @AWS_ET_JSON_STAGE,  
lateral flatten( input => parse_json($1))
```

```
SELECT $1 FROM @AWS_ET_JSON_STAGE;
```

```
SELECT $1 FROM @AWS_ET_JSON_STAGE;
```

```
SELECT DISTINCT METADATA$FILENAME FROM EXT_EMPLOYEES;
```

```
SELECT * FROM EXT_EMPLOYEES_JSON;
```

```
SELECT  
ARRAY_AGG(OBJECT_CONSTRUCT('DEPARTMENT_ID',D.DEPARTMENT_ID,'EMPLOYEE_  
ID',E.EMPLOYEE_ID,'EMPLOYEE_NAME',E.FIRST_NAME))  
FROM DEPARTMENTS D,EMPLOYEES E  
WHERE D.DEPARTMENT_ID=E.DEPARTMENT_ID
```

```
SELECT ARRAY_AGG(OBJECT_CONSTRUCT_KEEP_NULL(*))  
FROM EMPLOYEES  
group by DEPARTMENT_ID;
```

```
SELECT COUNT(*),DEPARTMENT_ID FROM EMPLOYEES  
GROUP BY DEPARTMENT_ID;
```

```
SELECT * FROM @AWS_ET_JSON_STAGE,  
lateral flatten( input => parse_json($1));
```

```
{  
  "COMMISSION_PCT": null,  
  "DEPARTMENT_ID": 100,  
  "EMAIL": "NGREENBE",  
  "EMPLOYEE_ID": 108,
```

```
"FIRST_NAME": "NANCY",  
"HIRE_DATE": "2002-08-17",  
"JOB_ID": "FI_MGR",  
"LAST_NAME": "GREENBERG",  
"MANAGER_ID": 101,  
"PHONE_NUMBER": "515.124.4569",  
"SALARY": 12008  
},
```

```
SELECT $1[0].EMPLOYEE_ID FROM @AWS_ET_JSON_STAGE
```

```
SELECT $1[0].EMPLOYEE_ID FROM @AWS_ET_JSON_STAGE
```

```
CREATE OR REPLACE EXTERNAL TABLE EXT_EMP_DEPT_JSON  
(  
  EMP_DEPT_JSON_DATA VARIANT AS (VALUE::VARIANT),  
  EMP_ID VARCHAR AS ($1[0].EMPLOYEE_ID::VARCHAR)  
)  
WITH LOCATION = @AWS_ET_JSON_STAGE  
FILE_FORMAT = JSON_ET_FILEFORMAT;
```